PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

- Team Name – Syntax Error
- Team Members -- Sharvesh J B - CB.EN.U4CCE26149
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- Selected Domain – Food Security, Agriculture & Rural Transformation
- Problem Statement – 86% of Indian farmers are small and marginal, but only 42.2% of farm households had access to technical advice. Most of them choose what to sow without knowing their soil health, climate conditions, pest risks, market demand, or likely price. The result is surplus crops, losses worth about 1.5 lakh crore a year after harvest, poor selling prices, and debt in 50.2% of farm households.

DOMAIN PROBLEM IN OUR WORDS

❖ ***The problem in our words***

In India, many farmers own 2 acres or less. Many of them do not know what their soil is good for, so they grow what their neighbour grows or what they grew the last year. If the whole village does this, everyone ends up with the same crop at the same time. This brings a price crash. Many of the vegetables rot before they reach a buyer, and they sell whatever is left for less than the cost of it, then they borrow to start the next season and the cycle repeats.

This is not a rare case. In India, more than 80 percent of farmers are small or marginal and half of all farmer households are in debt.

❖ ***What needs to be solved***

We need to fix three main problems.

1. The problem is not just in their soil; the pest and nutrient problems are noticed very late.
2. They don't know what to sow; nobody helps them look at their land type, the weather, the expected price, and how much the market needs.
3. The farmers don't have the knowledge of how much to grow or when to sell, so they overproduce, waste crops, and sell at a low price.

❖ ***Why it matters now***

The real scenario today is that from 2019 to 2022, agricultural production has increased by 23%. But the farmers grow more and still lose money. This happens entirely because of a lack of proper technical advice.

Now phones are everywhere, so a farmer can point a camera at a leaf or a patch of soil and get an answer right there. That wasn't possible a few years ago. In today's world, everything is possible and we are proposing this idea.

WHO IS AFFECTED?

Target users, stakeholders, beneficiaries

- Primary stakeholders
- Small and marginal farmers: face uncertain returns, potential crop surplus and difficulty finding buyers.
- Farmer Producer Organizations (FPOs): need to coordinate production and aggregate produce for sale.
- Buyers and retailers: face uncertainty in the quantity, quality and timing of available produce.
- Consumers: may face higher prices even while some produce goes unsold elsewhere.
- Transport and storage providers: need timely coordination to handle perishable harvests.



Credits : Amit Sengupta, Facebook



Credits : Farmlovers, Hyderabad

CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

- Crop selection : Farmers make planting decisions using available advice, experience, input costs and market information, which may be incomplete.
- Cultivation : Weather, water availability, pests and soil conditions affect crop growth and final yield.
- Harvesting : Actual production and harvest timing may differ from expectations. Multiple farms may harvest the same crop simultaneously.
- Selling : If buyers, transport or storage are unavailable, produce may remain unsold, lose value or spoil.



Credits : mongobay.com

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

- **Main difficulties and bottlenecks**
  - Farmers lack reliable information about crop suitability, weather conditions, and market demand.
  - Many farmers grow the same crops, leading to simultaneous harvests and potential oversupply.
  - Limited storage, transportation, and buyer access cause post-harvest losses.
- **Root causes behind them**
  - Limited access to timely agricultural advice and market information.
  - Crop planning is not sufficiently connected to expected demand and prices.
  - Inadequate post-harvest infrastructure and market coordination.
- **Consequences of leaving them unsolved**
  - Surplus production and avoidable food wastage.
  - Low selling prices and reduced farmer income.
  - Increased financial stress and risk of debt for farmers.



Credit: Subhash Sharma · The National

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

Solution	What it does well	Where it falls short
1. **Bharat-VISTAAR**	AI-driven, multilingual crop, weather & scheme advisories. Voice-first, strong pre-planting & growing-stage support.	Strong on advice, weak on linking planting decisions to actual expected harvest volume and post-harvest buyer/logistics coordination.
2. **e-NAM**	Digital marketplace for transparent selling & price discovery.	Mostly post-harvest. Very limited inter-state trade, infrastructure gaps, and almost no connection to pre-planting planning or surplus prediction.
3. **Kisan Sarathi**	Expert advice from KVKs + schemes + weather via app/call.	Reactive (query-based). Does not proactively connect the full cycle: planting decision → expected supply → buyer matching + logistics.

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

- **Hardware:** Soil-moisture sensor, soil-temperature sensor, suitable controller and connectivity module.
- **Software & Data:** AI decision-support system, satellite imagery, weather forecasts, crop information and agricultural market-price data.
- **Other Resources:** Agricultural experts, farmer groups, buyers, storage providers and transport partners.

Opportunity Areas

1. Demand-Aware Crop Planning
Match crop suitability, costs, weather and market demand. Flag potential oversupply before planting.
2. Soil & Crop Intelligence
Monitor soil moisture and temperature. Combine soil tests, satellite and weather data.
3. Harvest & Waste Prevention
Estimate yield and harvest timing. Connect surplus to buyers, storage and transport.
4. Visual AI Assistant
Identify crop-health symptoms and agricultural products from images.

INITIAL DIRECTION & EXPECTED OUTCOME

What Is Your Expected Outcome?

INITIAL DIRECTION :

Demand-Aware Farming Intelligence

- **Analyze:** Combine satellite data, soil information, weather and market trends.
- **Recommend:** Suggest suitable crops based on cultivation costs, expected returns and demand.
- **Predict:** Estimate harvest quantities and identify potential surplus early.
- **Connect:** Link farmers with buyers, storage and transport to reduce food wastage.

EXPECTED OUTCOMES :

- Reduced avoidable food surplus and post-harvest losses.
- Better-informed crop selection and resource use.
- Improved market access and price transparency.
- More sustainable farming and financial decision-making.



Credits : Food and Agriculture Organization of the United Nations

TEAM CONTRIBUTION

Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Akhilesh Aravind J	Problem & Stakeholder Research	Analyze the problem, affected stakeholders and supporting evidence.
Gaurav T	Existing Solutions Research	Research existing agricultural platforms and identify the innovation gap.
Roahith P	Solution Design	Define the core concept and expected outcomes.
Sharvesh JB	Challenge Analysis	Identify key challenges, root causes and consequences.